

TS Question Paper 2016
Class 12 Chemistry, Paper-II

Time: 3 Hours

Max.Marks:60

Note: Read the following instructions carefully:

- 1) Answer all the questions of Section-‘A’. Answer any six questions from Section-‘B’ and any two questions from Section-‘C’.**
- 2) In Section-‘A’, Questions from Sr. Nos. 1 to 10 are of “very short answer type”. Each question carries two marks. Every answer may be limited to two to three sentences. Answer all the questions at one place in the same order.**
- 3) In Section-‘B’, Questions from Sr. Nos. 11 to 18 are of “Short answer type”. Each questions carries four marks. Every answer may be limited to 75 words.**
- 4) In Section-‘C’, Questions from Sr.Nos.19 to 21 are of” Long answer type”. Each questions carries eight marks. Every answer may be limited to 300 words.**
- 5) Draw labelled diagram, wherever necessary for questions in section-‘B’ and ‘C’.**

SECTION- A **$10 \times 2 = 20$**

Note: Answer all the questions:

1. What are isotonic solution? Give an example.

Ans:- An Isotonic solution refers to a solution that has equal solute concentrations across a semipermeable membrane.

Osmotic pressure is also same in both solutions across semipermeable membrane.

Example of isotonic solution is - Cleaning solution of contact lenses.

2. A reaction has a half-life of 10 minutes. Calculate the rate constant for the first order reaction.

Ans:- To find the rate constant, we use the formula:

$$k = \frac{0.693}{t_{1/2}}$$

$$k = \frac{0.693}{10}$$

$$K = 0.0693/\text{minutes}$$

3. What is the role of cryolite in the metallurgy of aluminium?

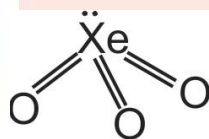
Ans:- Role of cryolite in the metallurgy of aluminium are

Firstly it helps to increase the electrical conductivity of the mixture. Cryolite is also helps in the lowering the melting point of the mixture.

4. Mention the shape and draw a diagram of XeO_3 .

Ans:- In XeO_3 , Xe exhibits sp_3 hybridisation. One of the hybrid orbitals is occupied by a lone pair, while other three are involved in the bond formation. This leads to a trigonal pyramidal structure of XeO_3 .

The shape of XeO_3 is triagonal pyramidal.



5. PH_3 has lower boiling point than NH_3 . Why?

Ans:- PH_3 has lower boiling point than NH_3 This is due to the fact that NH_3 has Hydrogen Bonding which increases the bond strength & thus increases the Boiling Point, while in case of PH_3 , no hydrogen bonding is possible

The boiling point of PH_3 is lower than NH_3 because the hydrogen bonding in NH_3 is stronger than PH_3 .

6. Aqueous Cu^{+2} ions are blue in color, whereas aqueous Zn^{+2} ions are colourless. Why?

Ans:- Configuration of $\text{Cu}_2 \pm [\text{Ar}] 3d^9$

Having an unpaired electron makes a F center which imparts colour to it.

Configuration of $\text{Zn}_2 \pm [\text{Ar}] 3d^{10}$

As it is not having any unpaired electron that's why it is colourless.

7. What is vulcanization of rubber?

Ans:- Vulcanization of rubber is a process of improvement of the rubber elasticity and strength by heating it in the presence of sulfur, which results in three-dimensional cross-linking of the chain rubber molecules (polyisoprene) bonded to each other by sulfur atoms

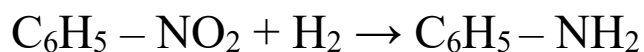
8. What is PHBV? How is it useful to man?

Ans:- PHBV is a copolymer. It is polyhydroxy butyrate -CO- β -hydroxy valerate. The monomers used are 3-hydroxy butanoic acid and 3-hydroxy pentanoic acid. It has an ester linkage. It is useful to mankind because it is biodegradable polymer and thus prevents environmental pollution.

9. How Aniline is obtained from Nitrobenzene?

Ans:- Aniline is prepared commercially by the catalytic hydrogenation of nitrobenzene or by the action of ammonia on chlorobenzene. The reduction of nitrobenzene can also be carried out with iron borings in aqueous acid. A primary

aromatic amine, aniline is a weak base and forms salts with mineral acids.



10. Write about carbylamines reaction.

Ans:-.In this reaction, the analyte is heated with alcoholic potassium hydroxide and chloroform. If a primary amine is present, the isocyanide (carbylamine) is formed which are foul smelling substances.

For example, the reaction with ethylamine:

SECTION- B

 $(6 \times 4 = 24)$

Note: Answer Any Six questions.

11. State Raoult's law. Calculate the mass of non-volatile solute (Molar mass 40 gr/mole-1) which should be dissolved in 114gr of octane to reduce its vapour pressure to 80%.

Ans:- Raoult's law states that the partial vapor pressure of each component of an ideal mixture of liquids is equal to the vapour pressure of the pure component multiplied by its mole fraction in the mixture.

Let the vapour pressure of pure octane be p_{10} .

Then, the vapour pressure of the octane after dissolving the non-volatile solute is $80/100 p_{10} = 0.8 p_{10}$.

Molar mass of solute, $M_2 = 40 \text{ g mol}^{-1}$

Mass of octane, $w_1 = 114 \text{ g}$

Molar mass of octane, (C_8H_{18}), $M_1 = 8 \times 12 + 18 \times 1 = 114 \text{ g mol}^{-1}$

Applying the relation,

$$(p_{10} - p_1) / p_{10} = (w_2 \times M_1) / (M_2 \times w_1)$$

$$\Rightarrow (p_{10} - 0.8 p_{10}) / p_{10} = (w_2 \times 114) / (40 \times 114)$$

$$\Rightarrow 0.2 p_{10} / p_{10} = w_2 / 40$$

$$\Rightarrow 0.2 = w_2 / 40$$

$$\Rightarrow w_2 = 8 \text{ g}$$

Hence, the required mass of the solute is 8 g.

12. Derive Bragg's equation.

Ans:- The phases of the beams coincide when the incident angle equals reflecting angle. The rays of the incident beam are in phase and parallel upto point z, which is the point at which top beam strikes the top layer. The second beam passes to next layer and is scattered by B. The second beam travels extra distance $AB + BC$. This extra distance is an integral multiple of the wavelength.

$$n\lambda = AB + BC.$$

But

$$AB = BC$$

$$n\lambda = 2AB \dots (1)$$

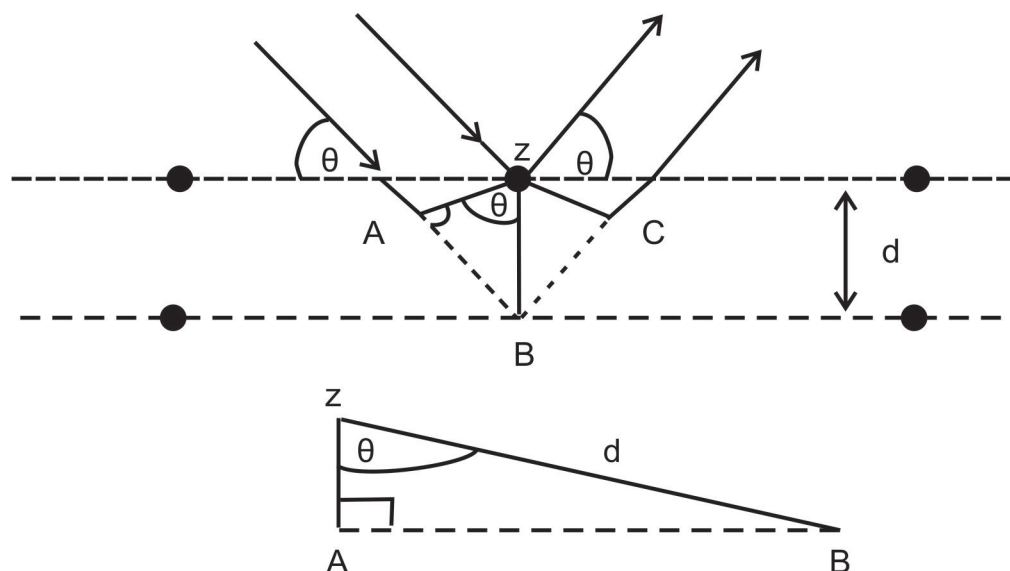
d is the hypotenuse of the right triangle Abz. Ab is opposite to angle θ

$$AB = d \sin \theta \dots (2)$$

Substitute equation (2) in equation (1)

$$n\lambda = 2d \sin \theta$$

This is equation for Bragg's law



13. What are emulsions? How they are classified? Give one example of each.

Ans:- Emulsion:- A colloidal system in which one liquid is dispersed in another liquid immiscible with it is called an emulsion.

The emulsions are classified into following two types.

(i) Emulsion of oil in water:

The disperse phase is an oil and the dispersion medium is water. For example, milk containing particles of liquid fat dispersed in water.

(ii) Emulsion of water in oil:

The disperse phase is water and the dispersion medium is an oil. For example, cod liver oil containing water particles dispersed

in oil.

14. Define calcinations and roasting. Give one example of each.

Ans:- Roasting is a process of heating the concentrated ore to a high temperature in presence of air. Generally, sulfide ores are roasted.

Example: $2\text{ZnS} + 3\text{O}_2 \rightarrow 2\text{ZnO} + 2\text{SO}_2$

Calcination is a process of heating ore in the absence of air to a high temperature but insufficient to melt the ore. Generally, carbonates and hydrated ores are calcined.

Ex: $\text{ZnCO}_3 \rightarrow \text{ZnO} + \text{CO}_2$

15. Write IUPAC names of the following coordination compounds.

a) $[\text{Co}(\text{NH}_3)_6]\text{Cl}_3$

b) $\text{K}_3[\text{Fe}(\text{CN})_6]$

c) $\text{K}_2[\text{Pd}(\text{Cl})_4]$

d) $[\text{Ni}(\text{Co})_4]$

Ans:- a) Hexaamminecobalt(III) chloride

b) Potassium ferricyanide

c) Potassium tetrachloroplatinate(II)

d) Nickel carbonyl

16. a) What is denaturation of proteins?

Ans:- Denaturation of proteins involves the disruption and

possible destruction of both the secondary and tertiary structures. Since denaturation reactions are not strong enough to break the peptide bonds, the primary structure (sequence of amino acids) remains the same after a denaturation process. Denaturation disrupts the normal alpha-helix and beta sheets in a protein and uncoils it into a random shape.

b) What are essential Amino acids? Give one example.

Ans:-An essential amino acid, or indispensable amino acid, is an amino acid that cannot be synthesized de novo by the organism at a rate commensurate with its demand, and thus must be supplied in its diet.

The amino acids which cannot be synthesized in the body and must be obtained through diet is known as essential amino acids.

Ex- Leucine

17. What are artificial sweetening agents and food preservatives? Give one example of each.

Ans:- Artificial sweetening agents:

These are chemicals that do not occur in nature but are synthesized in the laboratory. They have sweet taste but no food value.

Examples include saccharin, aspartame, sucralose and alitame.

Food preservatives:

These are substances, when added to food, are capable of

inhibiting, retarding or arresting, the process of fermentation, acidification or other decomposition of food by growth of microbes.

Examples include salt, sugar, vinegar, sodium benzoate, salts of sorbic acid and propionic acid etc.

18. Explain SN_1 And SN_2 reactions.

Ans: The SN_1 reaction is a substitution reaction in organic chemistry. "SN" stands for nucleophilic substitution and the "1" represents the fact that the rate-determining step is unimolecular. Thus, the rate equation is often shown as having first-order dependence on electrophile and zero-order dependence on nucleophile...

EXAMPLE – THE REACTION OF HBr with a Tertiary alcohol.

SN_2 indicates a substitution, nucleophilic, bimolecular reaction, described by the expression $\text{rate} = k [\text{Nu}][\text{R-LG}]$. This implies that the rate determining step involves an interaction between two species, the nucleophile and the organic substrate..

EXAMPLE – NaI with primary alkyl halides .

SECTION – C**(2 × 18 = 16)**

Note : Answer ANY TWO questions.

19. a) Define Kohlraush's law of independent migration of ions. Give its applications.

Ans:- Kohlrausch law of independent migration of ions: The law states that limiting molar conductivity of an electrolyte can be represented as the sum of the individual contributions of the anion and cation of the electrolyte.

Uses of Kohlrausch's law

- Calculation of Degree of dissociation
- Calculation of solubility of sparingly soluble salt
- Calculation of Dissociation Constant for weak electrolytes
- Calculation of Molar Conductivity for weak electrolytes at infinite dilution

Applications:-

- i) Determination of weak electrolytes.
- ii) Determination of the degree of ionisation of a weak electrolyte.
- iii) Determination of the ionisation constant of a weak electrolyte.
- iv) Determination of the solubility of a sparingly soluble salt.

19. (b) Describe the salient features of the Collision theory of

reaction rates of bimolecular reactions.

Ans:- Collision theory, theory used to predict the rates of chemical reactions, particularly for gases. The collision theory is based on the assumption that for a reaction to occur it is necessary for the reacting species (atoms or molecules) to come together or collide with one another. According to the collision theory, the rate at which a chemical reaction proceeds is equal to the frequency of effective collisions. Because atomic or molecular frequencies of collisions can be calculated with some degree of accuracy only for gases (by application of the kinetic theory), the application of the collision theory is limited to gas-phase reactions.

The salient features of the collision theory of reaction rates are:-

1. The reactants must collide with each other.
2. The molecules must have sufficient energy to initiate the reaction (called activation energy).
3. The molecules must have the proper orientation.

20. a) How is chlorine prepared in Deacon's process? How does it react with the following?

- i) Cold and dilute NaOH
- ii) Hot and concentrated NaOH

Ans:- Chlorine is prepared by oxidation of hydrogen chloride gas by atmospheric oxygen in the presence of CuCl_2 (Catalyst)

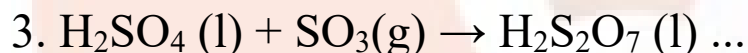
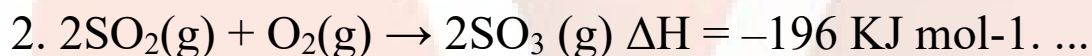
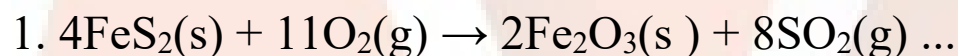
at 723K. This method is called Deacon's process.

i) Chlorine reacts with cold and dilute NaOH to produce sodium chloride, sodium chlorate(i) and water.

ii) Chlorine reacts with hot and concentrated NaOH to produce sodium chlorate (V).

b) Give chemical equations to manufacture of Sulphuric Acid by Contact Process.

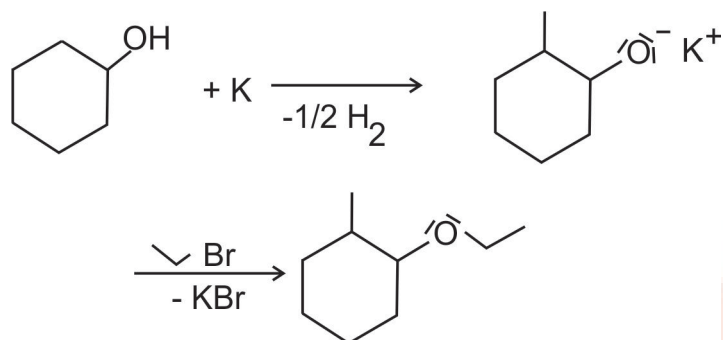
Ans:- $S(s) + O_2(g) \rightarrow SO_2(g)$ Sulfur dioxide can also be obtained by heating sulfide ores like pyrite in an excess of air:



21. Explain the following with one example.

a) Williamson's Synthesis

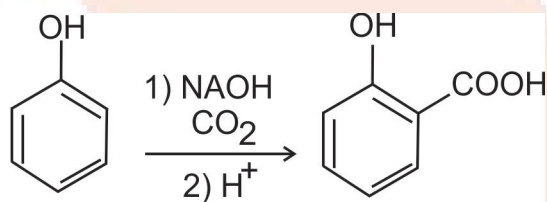
Ans:-The Williamson ether synthesis is the most widely used method to produce ethers. It occurs by an SN_2 reaction in which a metal alkoxide displaces a halide ion from an alkyl halide. The alkoxide ion is prepared by the reaction of an alcohol with a strong base such as sodium hydride.



b) Kolbe's reaction

Ans:-Definition of Kolbe reaction. 1 : the synthesis of a hydrocarbon (as ethane) by the electrolysis of a salt (as sodium acetate) 2 : the synthesis of salicylic acid by heating a mixture of sodium phenoxide and carbon dioxide under pressure at 180° to 200° C.

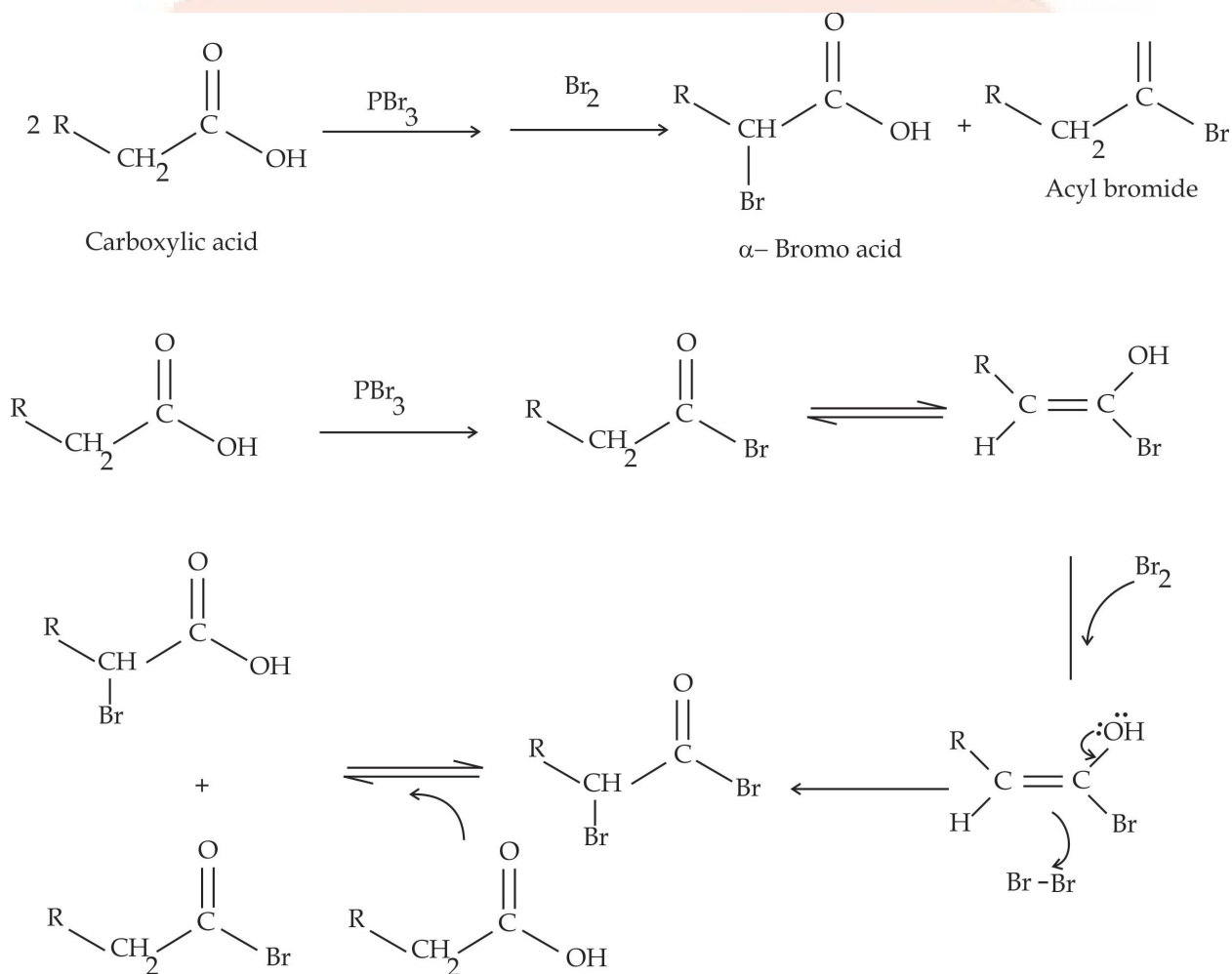
A base-promoted carboxylation of phenols that allows the synthesis of salicylic acid derivatives.



c) Hell-Volhard – Zelinsky reaction

Ans:- In the Hell Volhard Zelinsky reaction PBr_3 is used to replace the carboxylic OH with a bromide, resulting in a

carboxylic acid bromide. The acyl bromide can then tautomerize to an enol. This enol is then made to react with Br_2 at the α position forming an α -bromo acyl bromide. Reaction of the α -bromo acyl bromide with the original carboxylic acid yields the α -bromo carboxylic acid product and regenerates the acyl bromide intermediate.



d) Aldol condensation

Ans:- : An aldol condensation is a condensation reaction in organic chemistry in which an enol or an enolate ion reacts with a carbonyl compound to form a β -hydroxyaldehyde or β -hydroxyketone, followed by dehydration to give a conjugated enone.

Aldol condensations are important in organic synthesis, because they provide a good way to form carbon-carbon bonds. For example, the Robinson annulation reaction sequence features an aldol condensation; the Wieland-Miescher ketone product is an important starting material for many organic syntheses. Aldol condensations are also commonly discussed in university level organic chemistry classes as a good bond-forming reaction that demonstrates important reaction mechanisms.

